# Repository Town selection: comparative public water-data basis

**Decision date:** 25 October 2026
**Recommendation:** Select **Orange, NSW** as the single base town if the deciding criterion is the strongest publicly evidenced water-data basis. Retain Menindee only as a separate electrical-case study or comparative context; do not present Menindee and Broken Hill as one town. This is a comparative recommendation based on six supplied town audits, not an exhaustive statewide census or a claim that Orange has the most open operational telemetry.

## Executive finding

Orange has the strongest overall public evidence among the six candidates because it combines multiple town-attributed, recurrent source-operation and environmental-monitoring publications with drinking-water, supply planning, stormwater, sewerage and reuse documentation. Especially differentiating are quarterly Decision Support Tool (DST) reports, monthly source-extraction reports, and annual hydrology, aquatic-environment and groundwater monitoring reports, alongside dated drinking-water results and asset-management plans. The evidence is unusually broad and regularly refreshed for Orange, though much of it remains PDF/HTML rather than structured feeds. [1–5]

This is a recommendation for **best evidence base**, not a finding of best data openness in every dimension. No candidate demonstrated a complete, openly licensed utility asset GIS, town-wide SCADA/telemetry, normalized drinking-water laboratory history, or comprehensive wastewater/reuse operating series. Orange's lead is chiefly its breadth, town attribution and recurring operationally informative reporting—not a verified raw-data API or complete asset dataset.

## Rubric defined before comparative scoring

Scores are evidence-weighted, qualitative judgments on a 1–5 scale. A score of 1 means little directly evidenced public material in the dimension; 3 means useful but incomplete or mixed evidence; 5 means comparatively strong and directly evidenced coverage. Half-points distinguish close judgments. Scores describe the **public evidence located in the audits**, not underlying utility capability or absolute data completeness.

| Dimension | Weight | What is scored (and what is not) |
|---|---|---|

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|---|---|---|
| Public material breadth / source richness | 15% | Number and diversity of distinct relevant public source categories (supply, quality, security, wastewater, hydrology, etc.). Volume of documents alone does not establish operational value. |
| Town-specific operational utility | 30% | Direct attribution to the named town and usefulness for understanding measured operations, as distinct from general plans, proposed projects, utility-wide aggregates, or regional context. |
| Machine-readable spatial / hydrological access | 25% | Verified or credibly documented access to structured downloads, APIs, station observations or spatial services, with town/station relevance. General portal existence earns limited credit; a specific station/time series not verified is not treated as established. |
| Temporal depth and update | 20% | Recurrence, historical span and evidence of recent refresh for town-relevant observations/report series. Static plans and project announcements alone score lower. |
| Reuse-term clarity | 10% | Evidence of explicit dataset-level open terms. Public viewing or download without an identified licence is not counted as clearly reusable. |

**Calculation:** weighted score = sum of (dimension score × weight), on a 5-point scale. The result is rounded to one decimal and remains a comparative heuristic. Breadth is deliberately only 15% so a large stack of plans cannot outweigh weak operational specificity, machine access, or time depth. No separate “raw document count” score is used because the audits do not offer a uniform document census.

## Comparative scoring

| Candidate | Breadth (15%) | Town operational utility (30%) | Machine-readable spatial / hydro (25%) | Ten (20 |
|---|---|---|---|---|
| **Orange** | 4.5 | 4.0 | 3.0 | 4.5 |
| **Broken Hill** | 4.0 | 4.0 | 3.0 | 3.5 |
| **Menindee / Broken Hill corridor** | 4.0 | 3.5 | 3.0 | 4.0 |
| **Dubbo** | 3.5 | 3.5 | 3.0 | 3.0 |
| **Tamworth** | 3.5 | 3.5 | 2.5 | 3.0 |
| **Cobar–Nyngan** | 3.0 | 3.0 | 2.5 | 2.5 |

Scores are not statistically precise. In particular, access portals were not exhaustively queried station-by-station, licensing was not audited item-by-item, and the candidate scopes differ. A difference of a few tenths should not be read as a validated rank interval. The decimals make the rubric reproducible, not the evidence more certain.

## Why Orange leads

Orange's strongest comparative advantage is evidence across multiple water-system stages with repeated publication. Council's water-supply page links quarterly source-operation/DST reports, water-use summaries, annual raw-water/hydrology/aquatic-environment and groundwater reports, plus monthly source-extraction reports. These offer more recurring, town-attributed operational context than static strategy documents alone, even though publication as individual PDFs prevents treating them as a clean machine-readable time series. [1] [4]

The evidence also spans treatment quality (the 2024 Icely Road report with sample counts and summarized physical, chemical, microbiological and PFAS results), supply infrastructure and asset planning, stormwater-harvesting schemes, and sewerage/reuse planning. The Orange source material is broader and more temporally recurrent than the strongest alternatives, while explicitly identifying limitations: no verified Council asset GIS, no confirmed station-specific WaterNSW/BOM series for Orange, no raw sample-level drinking-water export, and no comprehensive wastewater/reuse or stormwater operating series. [2–6]

State hydrology portals and statewide spatial layers provide useful potential context but cannot be credited as confirmed Orange observations without station-level validation. The statewide NSW Foundation Spatial Data Framework Water theme is explicitly CC BY 4.0, but that does not license Council's reports or establish Orange utility network geometry. [7] [8]

## Strongest runner-up: Broken Hill

Broken Hill is the closest single-town challenger. Its town-specific quarterly and annual drinking-water reports extend through recent years; Essential Water also publishes wastewater-treatment monitoring and licence material with stated monthly Wills Street and quarterly South sampling, and town-specific supply documentation. BOM groundwater discovery/downloads and WaterNSW data/API paths add potentially useful machine access, although no definitive supply-relevant groundwater time series was validated and the API's documentation/testing requires registration. Much of the local utility evidence is PDF/web-based, and no complete local network GIS or broad town operational telemetry was verified. [9–13]

Broken Hill loses to Orange on the selected rubric because Orange's audited evidence has a stronger combination of recurrent source-operation reporting (quarterly and monthly), annual hydrology/ecology/groundwater reports, stormwater harvesting documentation and sewerage asset planning. Broken Hill's repeated drinking-water and wastewater compliance reporting is a real strength, but does not close the machine-readable and network-asset gaps, and its water-security research includes older investigation rather than a current operational groundwater series. The two are close enough that a future validation of machine-readable exports or local station history could narrow or change the ranking.

## Apples-to-apples limits and the combined-corridor issue

The candidates are not perfectly comparable units. Orange, Dubbo, Tamworth and Broken Hill are town-centered audits, while **Cobar–Nyngan** spans two councils and **Menindee / Broken Hill supply corridor** combines distinct settlements, suppliers and supply contexts. Comparing a two-town corridor's aggregate source inventory with one town can inflate breadth and temporal coverage. Conversely, a town-specific utility may report through a larger council or utility boundary, making town attribution weaker than its publication volume suggests.

The corridor must not be collapsed into a fictional single town: Essential Water's sources distinguish Menindee's Darling River supply from Broken Hill's Murray River supply via the Wentworth pipeline (following a 2019 source change). Menindee Lakes are relevant to regional resource and historical/security context, but their gauges and storages are not

direct measurement of Broken Hill's delivered treated water. Corridor-wide gauge observations and reports cannot be counted as town-network telemetry. [9] [10]

For this reason, the table scores the corridor conservatively on town operational utility, and the standalone Broken Hill score is shown separately rather than merged with it. A true combined regional case may be appropriate for a study of interconnected source security; it is not a like-for-like choice of one Repository Town.

## Recommendation on the existing Menindee electrical case

**Shift the single base town to Orange** if maximum public water-data evidence is the priority. Keep the existing Menindee electrical case intact as a separate electrical-focused case, or as a clearly labeled cross-case comparator; do not silently relabel it as Orange, and do not claim Menindee is the best water-data base on the present evidence. Orange and Menindee have different systems and evidence profiles, so this is a scope change, not a data merge.

If the repository cannot support two cases and continuity of the existing Menindee work is a hard constraint, the defensible fallback is to retain Menindee for the electrical case while explicitly accepting a weaker/uneven public water-data basis. The current audit does not support claiming Menindee alone beats Orange for water-data evidence. The recommendation prioritizes the stated maximum-public-water-data criterion over convenience or prior electrical-case continuity.

## Evidence limitations and validation before implementation

The underlying audits explicitly describe themselves as focused inventories, not exhaustive searches. This comparison inherits that boundary and the supplied source judgments; it has not independently re-extracted every portal, verified every PDF, or tested all downloads. Most important unresolved checks are: (1) whether Orange's periodic reports contain extractable, consistently defined underlying tables; (2) which Orange-area WaterNSW/BOM stations have relevant parameters and date coverage; (3) whether any candidate utility has since exposed operational data or asset GIS absent from the audits; and (4) dataset-specific licence and attribution terms. Those checks could affect machine-readability and reuse scoring. Proposed infrastructure is never treated here as commissioned operation. Groundwater/resource context is not conflated with town supply operations.

Before ingesting data, record the actual file/page version and observation period, custodian, geography, measurement unit, missingness/quality flags, and item-level reuse

terms. Treat published PDF summaries as citations/context unless data have been extracted and checked; do not infer continuous observations from periodic reports.

## References


1. Orange City Council, Orange' s water supply — DST archive, source-operation and monitoring entry point.
2. Orange City Council, Drinking Water Monitoring Report Orange City Council 2024.
3. Orange City Council, Water Supply Asset Management Plan 2020–2049.
4. Orange City Council, Macquarie River to Orange Pipeline reports and monthly extraction reports.
5. Orange City Council, Sewerage Asset Management Plan 2023–2052 and stormwater harvesting documentation.
6. Orange City Council, treatment plant and PFAS information; PRW demonstration-plant business-case tender (proposal-stage, not operating-data evidence).
7. NSW DCCEEW, water monitoring stations; WaterNSW, real-time water data; BOM, Water Data Online.
8. NSW Government, Foundation Spatial Data Framework Water theme and licensing.
9. Essential Water, service and supply description and water-quality reports.
10. Essential Water, wastewater treatment plant monitoring and EPL documents; IWCM strategy page.
11. WaterNSW, water data and API information; BOM, Australian Groundwater Explorer.
12. NSW DCCEEW, local water utility performance.
13. Geoscience Australia, Broken Hill Managed Aquifer Recharge project.
14. Dubbo Regional Council, Water Quality and Treatment; advanced wastewater plant completion notice.
15. Tamworth Regional Council, Water Supplies and Wastewater.
16. Cobar Shire Council, Water Supplies Asset Management Plan; Bogan Shire Council, drinking-water quality management report.
17. WaterNSW, Menindee Lakes; BOM Murray–Darling Basin Water Information Portal, Lower Darling.